

HW2 Instruction Manual
EE 134 Digital IC Design
Winter 2026
University of California, Riverside

Access to LTspice and Transistor Models

For this course, we will primarily use LTspice for circuit-level simulations.

To access the simulator, go to:

<https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>

Download the most recent version and follow the instructions to install on your computer.

For the transistors, we will be using a commercial library: GlobalFoundries 180nm process. It is also called a PDK (process design kit).

The library is provided as a zip file on Canvas, under xxx. Download it too, unzip it, and keep it on your computer. Inside this folder called gf180_pdk, you will see multiple .txt files, each of which models some devices like transistor or resistor in this PDK. For this course, we will use **cmos_tt.txt** unless otherwise specified. Processes with the suffix of _t stand for typical in the process corner. cmos_tt then means both the NMOS and PMOS are typical. _s and _f respectively standards for slow and fast, so the devices have slightly different characteristics.

Using LTspice

In LTspice, make sure you include the SPICE directive of “.inc xxx\gf180_pdk\cmos_tt.txt”, where xxx is the full directory path to your PDK folder on your computer. For each NMOS and PMOS device we use, remember to change the model name to **nmos_3p3** and **pmos_3p3** respectively.

Some good practices to avoid bugs:

- 1) Don't leave terminals floating.
- 2) Make sure terminals that should be grounded are indeed grounded.
- 3) Make sure body terminals are connected somewhere (by default, PMOS to VDD and NMOS to GND)
- 4) Initialize voltage level in each voltage source, even if you are sweeping it in DC analysis.

Some tutorials can be found in Canvas. Feel free to also search for other resources online for tutorial and debugging.

It may be more convenient to create a separate .asc file for each question that requires LTspice.

Part 1: CMOS Logic Gates

Q2.1: Design the following complex logic in CMOS style.

$$Y = A \cdot (B + C) + (D \cdot E)$$

Use LTspice to show the circuit schematic. Each input and output port should be clearly labeled. Show a screenshot of the schematic with labels. You may use the “label net” feature to specify the name and attach to the corresponding port.

Q2.2: Draw a CMOS XOR gate in LTspice. Assume that the inverted inputs are available. For this XOR gate, determine the widths of all PMOS and NMOS transistors (relative to the unit inverter) so that the worst-case equivalent resistance for pull-up equals the unit inverter pull-up resistance, and the worst-case equivalent resistance for pull-down equals the unit inverter pull-down resistance. Assume the lengths of all transistors are identical. Assume the unit inverter has PMOS width of 2 and NMOS width of 1. Annotate the sizing of each transistor on your LTspice schematic. No simulation is needed.

Q2.3: In LTspice, design a 2-input CMOS NOR gate. Each NMOS is sized at 2um/1um, and each PMOS is sized at 8um/1um. The NOR gate will drive a capacitor of 10fF. Empirically determine the input pattern that leads to the worst-case delay. For each input transition, you may model with PULSE(0 3 5n 1n 1n 8n 20n). Remember the delay is calculated from the input midpoint to the output midpoint.

Q2.4: You will estimate the switching power of a logic gate. Use the same circuit setup as in Q2.3: A 2-input NOR gate with NMOS (2um/1um) and PMOS (8um/1um), driving a 10fF capacitor. In LTspice, first run the transient for 1000 ns. After transient simulation, hold your left “alt” key and left-click on the VDD power supply (not the input PULSE). You should see a thermometer sign if you are hovering over the correct location. If done correctly, a plot of the power (represented by voltage * current), should appear. Hold the left “ctrl” key and click on the title of the plot (V(xxx)*I(xxx) where the xxx depends your node names). Report the average power in absolute value and integral (which represents energy). Next, redo the same exercise with the transient simulation running for 2000ns. Report the average power and energy. Discuss which parameter stays mostly constant across time and which parameter changes over time, and why.

Part 2: Other Logic Families

Q2.5: We will use the pass transistor to estimate the threshold voltage (Vth) drop. For NMOS, apply a DC gate voltage of 3V. Use NMOS transistor with 2um/1um. For the drain, apply a voltage pulse with PULSE(0 3 10n 1n 1n 18n 40n). On the source side, let the NMOS drive a 1fF load. Compare the ideal input voltage of 3V and the output voltage, and the voltage difference

towards the end of the pulse (steady-state) will be your V_{th} estimate. Report the estimated V_{th} value and show the waveform.

Q2.6: Repeat Q2.5 for PMOS. Use PMOS transistor with $4\mu\text{m}/1\mu\text{m}$. Remember the signs and that you will compare the output voltage with GND (0V) instead. Report the estimated V_{th} value and show the waveform.

Q2.7: In LTspice, design two versions of a 2:1 multiplexer. Multiplexer has the following logic.

$$F = A.S + B.\overline{S}$$

In the first version, you will use standard CMOS gates to implement this logic. In the second version, you will use transmission gates. For all NMOS, size as $2\mu\text{m}/1\mu\text{m}$. For all PMOS, size as $4\mu\text{m}/1\mu\text{m}$. Report the worst-case latency of each design.

Q2.8: For the two designs in Q2.7, measure and report the average power consumption of each (over some reasonable time duration like 1000 ns). Discuss why one consumes more power than the other.

Q2.9: In LTspice, design a 2-input NAND gate using dynamic logic style with footer. Size each NMOS to be $3\mu\text{m}/1\mu\text{m}$ and each PMOS to be $1\mu\text{m}/1\mu\text{m}$. You may use PULSE(0 3 5n 1n 1n 8n 20n) for both input data and clock. Show the circuit schematic. Measure and report the average power consumption (over some reasonable time duration like 1000 ns).

Q2.10: Based on the dynamic NAND gate you designed in Q2.9, use simulation to identify its switching pattern that leads to the worst-case delay. Compare this delay with the worst-case delay of a CMOS-based NAND gate with NMOS of $2\mu\text{m}/1\mu\text{m}$ and PMOS of $2\mu\text{m}/1\mu\text{m}$. Discuss if dynamic logic is superior in terms of speed.

Submission Instruction

All solutions (for both analytical questions and explanations of simulation results) must be typed. No handwriting. For any equations, you may use the "Equation" feature in Word.

The texts and waveforms in all screenshots of LTspice results must be readable.

Submit the report that documents each question's required items in a PDF file. The report should include your name. The PDF is named hw2_firstname_lastname.pdf. Upload the report to Canvas assignment.

Grading

Each question in this homework is worth 2 points (Q2.1 is worth 2 points, Q2.2 is worth 2 points etc.). Some questions have multiple small parts. Grading for each question is given in a ternary format: 0 (missing or entirely incorrect), 1 (partially correct with major errors or missing justification), or 2 points (correct approach with correct result and clear explanation).